PROJECT AURIA:
SOLAR POWERED BACKPACK

Authors: Meya Johnson[1], Kangi Ssemakula[2], Raphael Benjamin, Denisia Jocktane-Akagah, Isaac Ackun Jr., Chase Benjamin, Mufarowashe Musodza, and Susan Fru-Ndi

Rochester Institute of Technology | National Society of Black Engineers

Research Question

How can current solar backpacks be improved to provide more reliable, eco-friendly charging for students, and communities with limited power access?

Abstract

Reliable access to energy remains a persistent challenge for students and communities that lack consistent electrical infrastructure. Solar-powered backpacks offer a portable and sustainable solution by allowing users to generate energy while moving throughout their day. Despite this potential, many existing designs struggle to provide consistent performance due to limitations in energy efficiency, overheating, durability, and overall usability. These challenges reduce their effectiveness, particularly in environments where dependable charging is essential.

Advancements in photovoltaic technology, cooling systems, and smart materials present an opportunity to significantly improve the functionality of these devices. By analyzing current research and engineering developments, this study explores how solar backpacks can be redesigned to optimize energy capture, regulate heat, and better align with user needs. The goal is to develop a more reliable and environmentally sustainable solution that supports both everyday student use and broader applications in communities with limited access to power.

NSBE/ Environmental Solution(s)

Addressing the shortcomings of existing solar backpack systems requires a comprehensive approach that integrates technical innovation with practical design considerations. Enhancing photovoltaic efficiency remains a critical priority, as improved energy conversion enables more reliable performance under varying environmental conditions.

The incorporation of lightweight and flexible solar panels can further increase adaptability while maintaining functionality, allowing the system to better accommodate user movement and daily use. Effective thermal regulation is equally essential, as excessive heat negatively impacts both energy output and component longevity; implementing passive cooling mechanisms, such as optimized airflow structures and thermally conductive materials, can mitigate these effects without introducing unnecessary system complexity.

1

Table 1. Selection of solar backpack in the market.

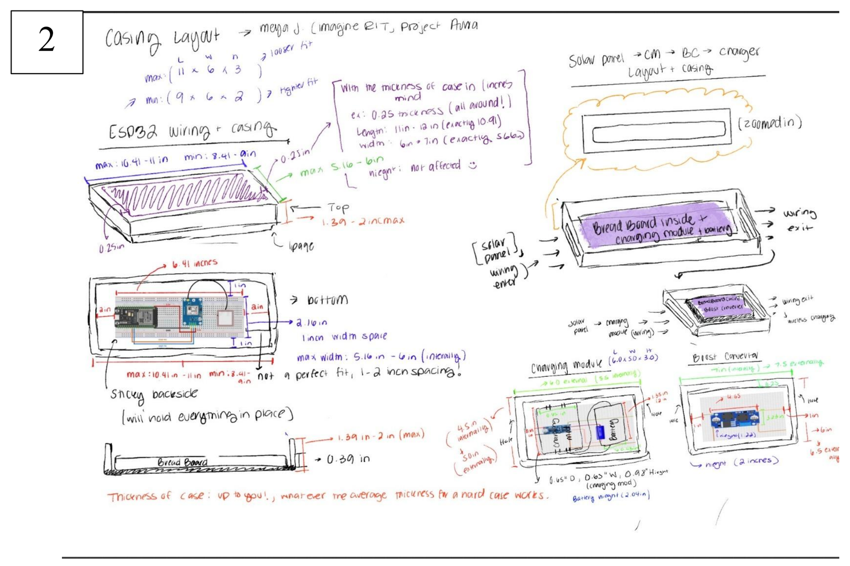
Sold By	Watts (W)	Voltage (V)	Battery Bank	Bag Size (L*W*H) (cm)	Weight (kg)	Price (£)
Vitechbiz [10]	6.5	6.0	No	35*22*52	1.30	90.00
TVCMALL [11]	6.5	6.0	No	55*36*23	1.35	40.40
ECEEN [12]	7.0	--	Yes (10AH)	65*60*30	2.00	139.99
Sunsky [13]	6.5	6.0	Yes	22*8.5*32	1.80	71.99
Glorysolar [14]	7.0	5.0	No	55*36*23	1.34	57.39

Picture 1: Materials list of the materials based on a manufactured solar backpack.

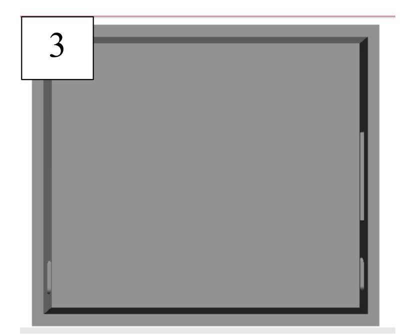
Material selection contributes to both durability and sustainability, with the integration of reinforced or adaptive fabrics enhancing structural integrity and user comfort. Emphasizing accessibility and usability ensures that these technological improvements translate into practical solutions, particularly for students and communities with limited access to reliable power.

Project Auria Procedure

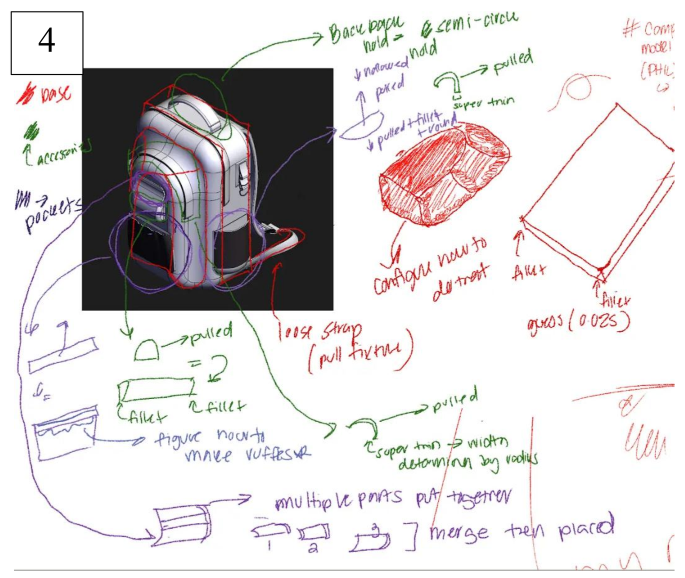
The design process began with a structured conceptualization phase focused on translating research findings into a functional and user-centered system. Initial sketching and modeling were conducted using both hand-drawn references and digital platforms, including Figma and CAD-based tools, to visualize the physical and electronic layout of the backpack. Emphasis was placed on spatial organization, ensuring that each component, including the solar panel, battery system, and embedded electronics, could be integrated without compromising comfort, weight distribution, or usability.



Picture 2: Sketches of the casing for the microelectronics ESP32 and IN219.

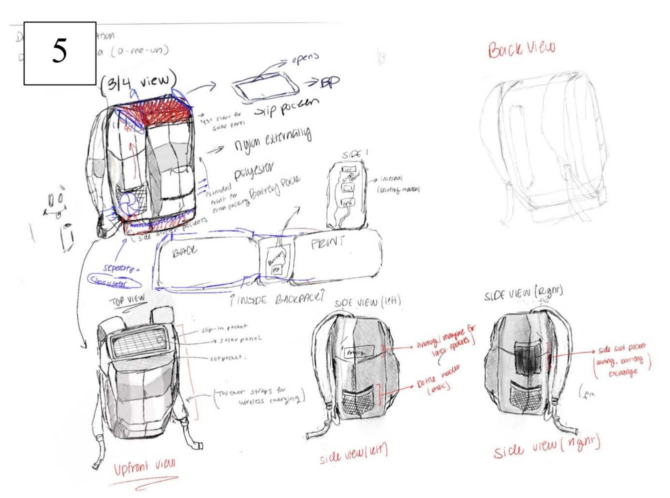


Picture 3: One of the CAD designs for the cases specifically the IN219.

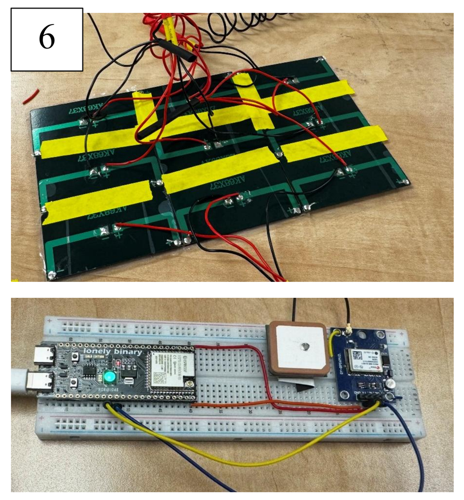


Picture 4: CAD (Fusion 360) drawing of ideas for our project.

Following the design phase, the project progressed into system construction and implementation. Component selection was guided by accessibility, cost-efficiency, and compatibility, incorporating a 5V flexible solar panel, TP4056 charging module, rechargeable battery pack, boost converter, and both wired and wireless charging outputs. Additional microcontroller integration included the use of an ESP32 system paired with an INA218 power sensor to monitor voltage and current flow throughout the system.



Picture 5: Sketches of our ideal backpack design and placements.



Picture 6: Solar panel parallel circuitry and the ESP32 GPS tracking builds.

Picture 7: The complete layout of the solar pack of where the microelectronics sit on the backpack.

Background

The increasing reliance on portable electronic devices has made access to consistent and reliable energy a fundamental part of daily life, particularly for students who depend on phones, laptops, and tablets for academic, professional, and social engagement. Despite this growing dependence, access to stable charging infrastructure is not always guaranteed, especially in outdoor environments, during travel, or within communities where electrical systems are limited or unreliable. Like how environmental challenges such as marine plastic pollution in Senegal are intensified by gaps in infrastructure, inefficient systems, and uneven resource distribution , limitations in energy accessibility can also be traced to inefficiencies in design and implementation.

Solar-powered backpacks have emerged as an innovative attempt to address this issue by integrating renewable energy into a portable, wearable system, allowing users to generate power throughout the day. However, current models often fall short of their intended purpose due to inconsistent energy output, overheating, limited battery storage, and lack of user-centered design considerations. These challenges reduce their effectiveness and prevent widespread adoption, particularly in contexts where reliability is essential.

Meet the Senators !

Meya Johnson
Astrophysics and Biotechnology

Kangi Ssemakulua
Packaging Science

Project Outline ->

Conclusion

The refinement of solar-powered backpack technology represents a significant opportunity to address persistent challenges related to energy accessibility and environmental sustainability. Improvements in efficiency, thermal management, material durability, and overall system integration enhance the practicality and reliability of these devices, allowing them to function effectively in diverse real-world conditions.

The incorporation of renewable energy into portable, everyday objects reflects a broader shift toward sustainable innovation, demonstrating how targeted design improvements can contribute to long-term environmental goals. For students, such advancements facilitate consistent access to power, supporting academic engagement and productivity, while for underserved communities, they provide a viable alternative to traditional infrastructure limitations. Continued exploration and development in this field hold the potential to expand access to clean energy solutions while promoting responsible engineering practices, ultimately contributing to a more sustainable and equitable global landscape.

References

Abdelmalek, Z., Rehman, K. U., Al-Mdallal, Q. M., Al-Kouz, W., & Malik, M. Y. (2020). Dynamics of thermally magnetized grooved flow field having uniformly heated circular cylinder: Finite element analysis. *Case Studies in Thermal Engineering, 21.* https://doi.org/10.1016/j.csite.2020.100718

Blondeau, J., Museur, T., Demaude, O., Allard, P., Turoni, F., & Mertens, J. (2020). Cost-effective flexibilization of an 80 MWe retrofitted biomass power plants: Improved combustion control dynamics using virtual air flow sensors. *Case Studies in Thermal Engineering, 21.* https://doi.org/10.1016/j.csite.2020.100680

Budiyanto, M. A., & Shinoda, T. (2020). An investigation of thermal stratification in refrigerated container yards. *Case Studies in Thermal Engineering, 21.* https://doi.org/10.1016/j.csite.2020.100703

Das, M., & Akpinar, E. K. (2020). Determination of thermal and drying performances of the solar air dryer with solar tracking system: Apple drying test. *Case Studies in Thermal Engineering, 21.* https://doi.org/10.1016/j.csite.2020.100731

Dwivedi, P., Sudhakar, K., Soni, A., Solomin, E., & Kirpichnikova, I. (2020). Advanced cooling techniques of P.V. modules: A state of art. *Case Studies in Thermal Engineering, 21*, 100674. https://doi.org/10.1016/J.CSITE.2020.100674

Acknowledgements

The development of this project would not have been possible without the support, collaboration, and contributions of numerous individuals who dedicated their time and effort throughout the design and research process. Special recognition is extended to Frances Mouangue, Wayne Jean Baptiste, Raphael Benjamin, Chimamanda Eze, Isaac Ackun Jr., Chase Benjamin, Joshua Vail, Lamont Black, Mailu Musila, Susan Fru-Ndi, Kingsley Nwabeke, Jayson Packer, CJ Scott, Tewan Ayong, and Denisia Jocktane-Akagah for their active participation, technical input, and continued engagement in advancing the project.

Additional appreciation is extended to all members of the Imagine RIT organization and NSBE community whose feedback, collaboration, and shared commitment to innovation helped shape the direction of this work. The collective effort and interdisciplinary perspective brought forward by these individuals reflect the importance of teamwork in addressing complex engineering challenges and advancing sustainable solutions.